

Reviewer Pack — Bourgain-Tzafriri Sub-Column Spectral Excess Bounds DISJ CC_...

Entry 14df14ed8261 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Bourgain-Tzafriri Sub-Column Spectral Excess Bounds DISJ CC_R

Entry ID: 14df14ed8261 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-17 22:43:48 UTC

1. Conjecture

Field A (mathematical branch): Bourgain-Tzafriri / Kashin restricted invertibility theory — average operator norm of random k -column submatrices of a centered sign matrix, in the Bourgain-Tzafriri 1987 / Kashin 1977 / Spielman-Srivastava 2010 tradition of restricted invertibility and column-subset selection. An arXiv search for ‘Bourgain Tzafriri’ AND (‘communication complexity’ OR ‘DISJOINTNESS’) returns 0 direct hits and <5 adjacent papers, all on Forster-style spectral sign-rank rather than sub-column dispersion. **Field B** (complexity object): Randomized two-party communication complexity $CC_R(M)$ of $N \times N$ Boolean matrices, with target M_DISJ_n on $N=2^n$ (Razborov 1992 $\Omega(n)$ regime), as an instance of unsaturated communication bounds via dispersion methods.

Statement:

Center $M \in \{0,1\}^{N \times N}$ to $\tilde{M} = 2M - J \in \{\pm 1\}^{N \times N}$. Fix $k = \lceil \log_2 N \rceil$ and let $S_1, \dots, S_{30} \subset [N]$ be uniformly random column subsets of size k drawn with seeds 1..30. Define the sub-column spectral excess $\xi(M) := (1/30) \cdot \sum_{s=1}^{30} (\|\tilde{M}|_{S_s}\|_{op}^2 / N - 1)$, where $\tilde{M}|_S$ is the $N \times k$ column submatrix. We conjecture that for every Boolean matrix M with $N \leq 32$: (i) $\xi(M) \geq 0$; (ii) $CC_R(M) \geq \lceil \log_2(1 + N \cdot \xi(M)/k) \rceil - 1$; (iii) $\xi(M_PARITY_n) \leq 0.05$ (so the bound gives $O(1)$, matching $CC_R = O(1)$) and $\xi(M_DISJ_n) \geq 0.5 \cdot k/n$ at $n=3,4,5$ (so the bound delivers $\Omega(\log N)$ for DISJ). A single seed-30 ensemble at any $n \in \{3,4,5\}$ with $\xi(M_DISJ_n) < 0.3 \cdot k/n$ or $\xi(M_PARITY_n) > 0.1$ falsifies the conjecture.

Rationale (proposer’s reasoning):

Sub-column spectral excess measures the average deviation of k -column Gram spectra from the orthogonal (Hadamard) isotropic baseline, where the baseline $\|\tilde{M}|_S\|_{op}^2 / N = 1$ holds exactly for parity-XOR matrices because their columns are orthogonal. Structured matrices like M_DISJ_n have correlated columns whose Gram matrix has a dominant ‘all-disjoint-tail’ eigenvector, forcing $\|\tilde{M}|_S\|_{op}^2 > N$ on most subsets. Crucially the invariant is RELATIONAL (an empirical spectral statistic over random column subsamples, not a single-number truth-table functional) and is not large in the natural-proofs sense (random Boolean matrices have $\xi \approx k/N \rightarrow 0$), so it sidesteps both the Razborov-Rudich barrier and the algebrization filter, while computationally reducing to 30 small SVDs.

Taxonomy category: DISPERSION_DISCREPANCY (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 5477c33731799c0f

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

Across 30 seeds per $n \in \{3, 4, 5\}$, mean $\xi(M_DISJ_n) \geq 0.5 \cdot k/n$ AND mean $\xi(M_PARITY_n) \leq 0.05$ AND mean $\xi(M_random) \leq 0.3 \cdot k/N$; per-seed predicted bound $\lceil \log_2(1 + N \cdot \xi/k) \rceil - 1$ must not exceed textbook $CC_R(DISJ_n) = \lceil n/4 \rceil$ or $CC_R(PARITY_n) = 2$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.90	SAFE	SAFE
NATURAL PROOFS	SAFE	0.78	SAFE	SAFE
ALGEBRIZATION	SAFE	0.82	SAFE	SAFE
KARP_LIPTON	SAFE	0.92	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 11 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - Bourgain Tzafriri restricted invertibility communication complexity lower bound - random column submatrix spectral norm sign matrix disjointness - Kashin column subset selection Boolean matrix communication complexity dispersion

Top relevant hits considered: - [<http://arxiv.org/abs/2208.05223v1>] Modular Bourgain-Tzafriri Restricted Invertibility Conjectures and Johnson-Lindenstrauss Flattening Conjecture - [<http://arxiv.org/abs/2503.20972v3>] p-adic Grothendieck Inequality, p-adic Johnson-Lindenstrauss Flattening and p-adic Bourgain-Tzafriri Restricted Invertibility - [<http://arxiv.org/abs/cs/0111062v2>] One-way communication complexity and the Neciporuk lower bound on formula size - [<http://arxiv.org/abs/1005.1005>] Supersymmetry in Random Matrix Theory - [<http://arxiv.org/abs/0904.1432v2>] Applications of random matrix theory to condensed matter and optical physics - [<http://arxiv.org/abs/1208.6015v1>] The spectral function of a first order elliptic system - [<http://arxiv.org/abs/2007.10307v2>] Optimal ℓ_1 Column Subset Selection and a Fast PTAS for Low Rank Approximation - [<http://arxiv.org/abs/2312.10602v1>] A Weighted K-Center Algorithm for Data Subset Selection

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=0, elapsed=0.3s

5.1 Generated Python source

```
import sys
import random
import math
import json

def matrix_multiply(A, B):
    n = len(A)
    m = len(B[0])
```

```

p = len(B)
result = [[0.0 for _ in range(m)] for _ in range(n)]
for i in range(n):
    for j in range(m):
        for k in range(p):
            result[i][j] += A[i][k] * B[k][j]
return result

def transpose(matrix):
    return [list(row) for row in zip(*matrix)]

def svd(matrix):
    n = len(matrix)
    m = len(matrix[0])
    A = [[float(x) for x in row] for row in matrix]
    ATA = matrix_multiply(transpose(A), A)
    s = [0.0] * min(n, m)
    for i in range(min(n, m)):
        if i >= len(ATA) or i >= len(ATA[0]):
            continue
        s[i] = math.sqrt(ATA[i][i]) if ATA[i][i] >= 0 else 0.0
    return None, s, None

def compute_spectral_excess(M_tilde, k, seed):
    random.seed(seed)
    N = len(M_tilde)
    xi = 0.0
    instances = 0
    for _ in range(30):
        S = sorted(random.sample(range(N), k))
        submatrix = [[M_tilde[i][j] for j in S] for i in range(N)]
        _, s, _ = svd(submatrix)
        if not s:
            continue
        op_norm_sq = s[0] ** 2 if s else 0.0
        xi += (op_norm_sq / N - 1)
        instances += 1
    xi /= 30 if instances > 0 else 1
    return xi, instances

def construct_disj_matrix(n):
    N = 2 ** n
    M = [[0] * N for _ in range(N)]
    for x in range(N):
        for y in range(N):
            M[x][y] = 1 if not (x & y) else 0
    return M

def construct_parity_matrix(n):
    N = 2 ** n
    M = [[0] * N for _ in range(N)]
    for x in range(N):
        for y in range(N):
            M[x][y] = (1 + (-1) ** bin(x & y).count('1')) // 2
    return M

```

```

def construct_random_matrix(n):
    N = 2 ** n
    M = [[0] * N for _ in range(N)]
    for x in range(N):
        for y in range(N):
            M[x][y] = random.randint(0, 1)
    return M

def center_matrix(M):
    N = len(M)
    row_sums = [sum(row) for row in M]
    col_sums = [sum(M[i][j] for i in range(N)) for j in range(N)]
    total_sum = sum(row_sums)
    M_tilde = [[0] * N for _ in range(N)]
    for i in range(N):
        for j in range(N):
            M_tilde[i][j] = 2 * M[i][j] - row_sums[i] - col_sums[j] + total_sum / N
    return M_tilde

def run_trial(seed):
    n_values = [3, 4, 5]
    results = []
    for n in n_values:
        N = 2 ** n
        k = math.ceil(math.log2(N))

        # Construct and center DISJ matrix
        M_disj = construct_disj_matrix(n)
        M_disj_tilde = center_matrix(M_disj)
        xi_disj, instances_disj = compute_spectral_excess(M_disj_tilde, k, seed)

        # Construct and center PARITY matrix
        M_parity = construct_parity_matrix(n)
        M_parity_tilde = center_matrix(M_parity)
        xi_parity, instances_parity = compute_spectral_excess(M_parity_tilde, k, seed)

        # Construct and center random matrix
        M_random = construct_random_matrix(n)
        M_random_tilde = center_matrix(M_random)
        xi_random, instances_random = compute_spectral_excess(M_random_tilde, k, seed)

        # Check conditions
        condition1 = xi_disj >= 0.5 * k / n
        condition2 = xi_parity <= 0.05
        condition3 = xi_random <= 0.3 * k / N

        conjecture_holds = condition1 and condition2 and condition3
        counterexample = ""
        if not condition1:
            counterexample = f"xi_disj={xi_disj} < 0.5*k/n={0.5*k/n}"
        elif not condition2:
            counterexample = f"xi_parity={xi_parity} > 0.05"
        elif not condition3:
            counterexample = f"xi_random={xi_random} > 0.3*k/N={0.3*k/N}"

        results.append({

```

```

        "n": n,
        "xi_disj": xi_disj,
        "xi_parity": xi_parity,
        "xi_random": xi_random,
        "instances_tested": instances_disj + instances_parity + instances_random,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    })

# Aggregate results
metric_values = [r["xi_disj"] for r in results]
metric_value = sum(metric_values) / len(metric_values) if metric_values else 0.0
instances_tested = sum(r["instances_tested"] for r in results)
conjecture_holds = all(r["conjecture_holds"] for r in results)
counterexample = next((r["counterexample"] for r in results if not r["conjecture_holds"]), "")

return {
    "metric_name": "xi_disj",
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97]
    trials = []
    for seed in seeds:
        result = run_trial(seed)
        trials.append({"seed": seed, **result})
    print(f"TRIAL: {json.dumps({'seed': seed, **result})}")

    metric_values = [trial["metric_value"] for trial in trials]
    mean_metric = sum(metric_values) / len(metric_values) if metric_values else 0.0
    std_metric = math.sqrt(sum((x - mean_metric) ** 2 for x in metric_values) / len(metric_values))
    support_fraction = sum(1 for trial in trials if trial["conjecture_holds"]) / len(trials) if trials else 0.0

    if all(trial["conjecture_holds"] for trial in trials):
        print(f"RESULT: SUPPORTED mean={mean_metric} std={std_metric} support_fraction={support_fraction}")
    elif any(not trial["conjecture_holds"] for trial in trials):
        first_failing_seed = next(trial["seed"] for trial in trials if not trial["conjecture_holds"])
        counterexample = next(trial["counterexample"] for trial in trials if not trial["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE reason=all_trials_failed")

```

6. Per-seed results

Seed	Metric value	Holds?	Counterexample
11	171.16764322916666	☐	xi_parity=25.05 > 0.05
23	135.80496961805557	☐	xi_parity=21.45 > 0.05
37	150.70896267361113	☐	xi_parity=22.35 > 0.05
53	95.11260850694445	☐	xi_parity=18.75 > 0.05
71	148.6762369791667	☐	xi_parity=23.25 > 0.05
89	137.0623480902778	☐	xi_parity=18.75 > 0.05

Seed	Metric value	Holds?	Counterexample
103	115.17371961805556	□	xi_parity=19.65 > 0.05
127	137.2179904513889	□	xi_parity=20.55 > 0.05
149	117.4211154513889	□	xi_parity=24.15 > 0.05
167	155.1325737847222	□	xi_parity=23.25 > 0.05
191	130.19533420138887	□	xi_parity=23.25 > 0.05
211	126.83040364583336	□	xi_parity=16.05 > 0.05
233	111.27076822916666	□	xi_parity=20.55 > 0.05
257	131.47354600694445	□	xi_parity=20.55 > 0.05
277	137.0428168402778	□	xi_parity=20.55 > 0.05
311	137.20427517361114	□	xi_parity=21.45 > 0.05
347	120.06495225694444	□	xi_parity=16.95 > 0.05
389	175.52319878472224	□	xi_parity=22.35 > 0.05
421	129.30965711805555	□	xi_parity=18.75 > 0.05
463	119.11555989583333	□	xi_parity=16.95 > 0.05
503	114.68014322916667	□	xi_parity=17.85 > 0.05
547	108.26382378472222	□	xi_parity=16.95 > 0.05
593	161.8962890625	□	xi_parity=24.15 > 0.05
631	109.49559461805556	□	xi_parity=16.95 > 0.05
677	95.28934461805557	□	xi_parity=20.55 > 0.05
727	132.66538628472222	□	xi_parity=20.55 > 0.05
773	137.6169487847222	□	xi_parity=18.75 > 0.05
821	112.67233072916667	□	xi_parity=18.75 > 0.05
877	133.46503906249998	□	xi_parity=14.25 > 0.05
929	139.62198350694447	□	xi_parity=19.65 > 0.05

Aggregate statistics:

Statistic	Value
n_seeds	30
metric_mean	130.90585214120372
metric_std	19.813464005215845
metric_ci95_half	7.234854118648917
metric_min	95.11260850694445
metric_max	175.52319878472224
support_fraction	0.0

7. Test stdout (last 2KB)

```
j", "metric_value": 119.11555989583333, "instances_tested": 270, "conjecture_holds": false, "counter
TRIAL: {"seed": 503, "metric_name": "xi_disj", "metric_value": 114.68014322916667, "instances_tested
TRIAL: {"seed": 547, "metric_name": "xi_disj", "metric_value": 108.26382378472222, "instances_tested
TRIAL: {"seed": 593, "metric_name": "xi_disj", "metric_value": 161.8962890625, "instances_tested": 2
TRIAL: {"seed": 631, "metric_name": "xi_disj", "metric_value": 109.49559461805556, "instances_tested
TRIAL: {"seed": 677, "metric_name": "xi_disj", "metric_value": 95.28934461805557, "instances_tested"
TRIAL: {"seed": 727, "metric_name": "xi_disj", "metric_value": 132.66538628472222, "instances_tested
TRIAL: {"seed": 773, "metric_name": "xi_disj", "metric_value": 137.6169487847222, "instances_tested"
TRIAL: {"seed": 821, "metric_name": "xi_disj", "metric_value": 112.67233072916667, "instances_tested
TRIAL: {"seed": 877, "metric_name": "xi_disj", "metric_value": 133.46503906249998, "instances_tested
TRIAL: {"seed": 929, "metric_name": "xi_disj", "metric_value": 139.62198350694447, "instances_tested
RESULT: FALSIFIED counterexample="xi_parity=25.05 > 0.05" first_failing_seed=11
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The empirical data directly falsifies the conjecture: support_fraction=0.0 across all 30 seeds, with every seed reporting xi_parity $\approx$ 18-25, vastly exceeding the 0.05 threshold required by clause (iii). The conjecture explicitly states ' $\xi(M_PARITY_n) \leq 0.05$ ' falsifies if exceeded, and every trial shows xi_parity > 0.1 . This is a clear metric-definition/construction issue — the PARITY matrix's centered version $\tilde{M}$ has all rows orthogonal up to sign, so any k-column submatrix has spectral norm² $\approx N$

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge falsified | original: The test's RESULT line reports FALSIFIED with counterexample xi_parity=25.05 > 0.05 on seed 11, and all 30 seeds show xi_parity in the 14-25 range, va | next: Replace raw spectral excess with a PARITY-insensitive functional (e.g., subtract the expected operator norm of a random ± 1 $N \times k$ matrix, $\approx (\sqrt{N} + \sqrt{k})^2$, or use the L1 $\rightarrow$ L2 sub-column norm) so that $\xi(PARITY) \approx 0$ while $\xi(DISJ)$ remains $\Omega(k/n)$.

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	claude_max	opus	0	0	333264
2	preregistration	claude_max	opus	0	0	7135
3	novelty	claude_max	opus	0	0	3747
4	novelty	claude_max	opus	0	0	10613
5	test_gen	mistral	codestral-latest	0	0	317048
6	test_gen	mistral	codestral-latest	0	0	314712
7	critic	claude_max	opus	0	0	9490
8	judge	claude_max	opus	0	0	8641

Totals: 0 input tokens, 0 output tokens, $\sim$ \$0.0000 cost, 1004650 ms total latency. Provider mix: {'claude_max': 6, 'mistral': 2}

(full prompt+response transcripts available in `research/audit/14df14ed8261.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/14df14ed8261.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/14df14ed8261.tar.gz` (if generated)